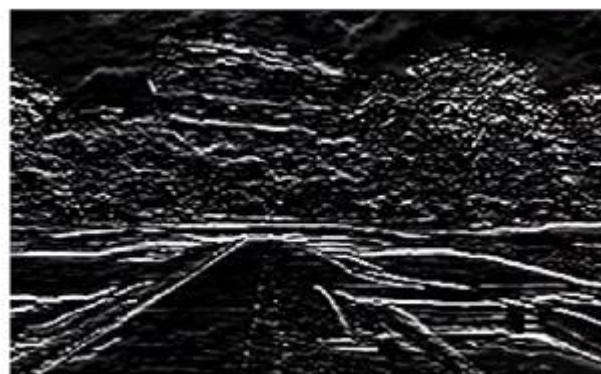
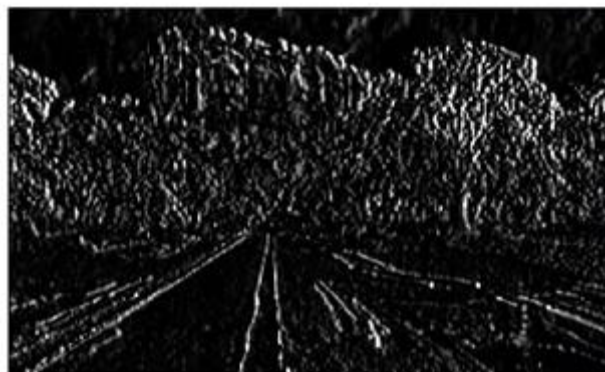


Spatial Filtering



Introduction

- We use spatial filtering to enhance images, e.g., noise filtering. We can use the same technique to locate objects in images, called template matching. After completing this lesson, we would be able to process images using filters, as done in popular image processing software such as Photoshop.
- Intensity transformations affect the brightness (intensity level) of an image. The resulting value of a pixel after an intensity operation is just dependent on the original value of the same pixel.
- In contrast, the resulting value of a pixel after a spatial filtering operation depends on the neighborhood of the pixel in question. So the space around the pixel in question matters, hence, the name spatial filtering.
- In linear spatial filtering we use the 2-D convolution operation.

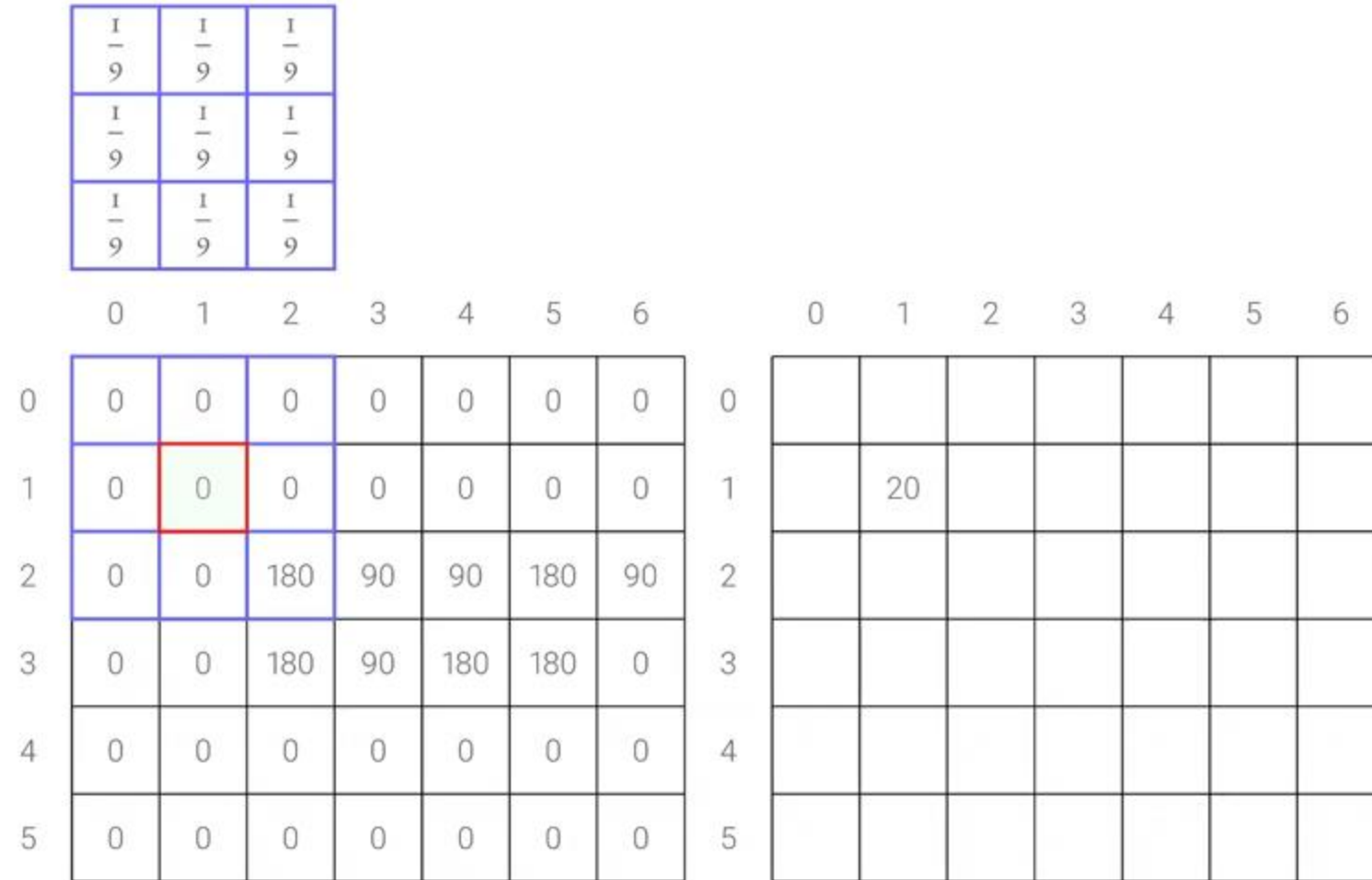


Figure: Convolution

filter
kernel
window

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Averaging

3x3

	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
2	0	0	180	90	90	180	90
3	0	0	180	90	180	180	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0

	0	1	2	3	4	5	6
0							
1		20					
2							
3							
4							
5							

Figure: Convolution

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
2	0	0	180	90	90	180	90
3	0	0	180	90	180	180	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0

	0	1	2	3	4	5	6
0							
1		20	30				
2							
3							
4							
5							

Figure: Convolution

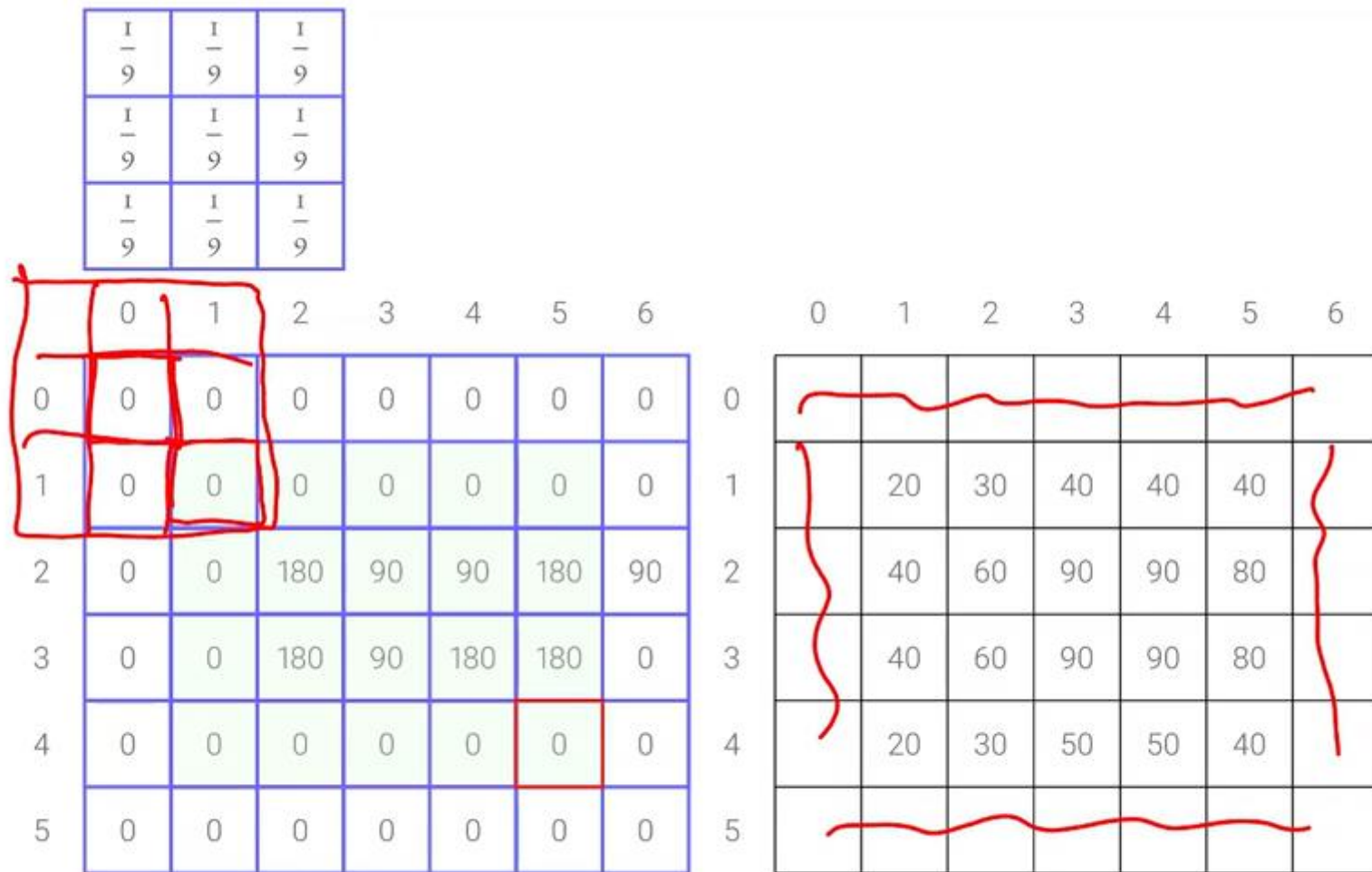


Figure: Convolution

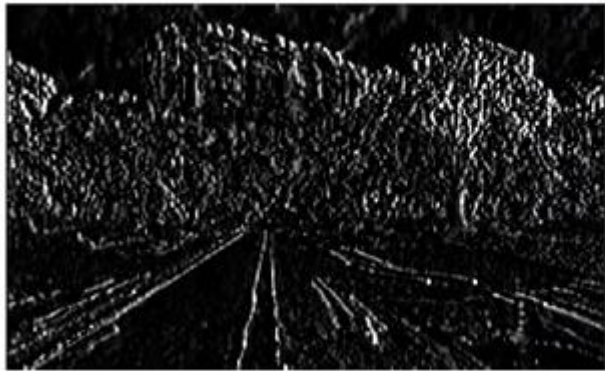
Examples of Effect of Kernel Choices



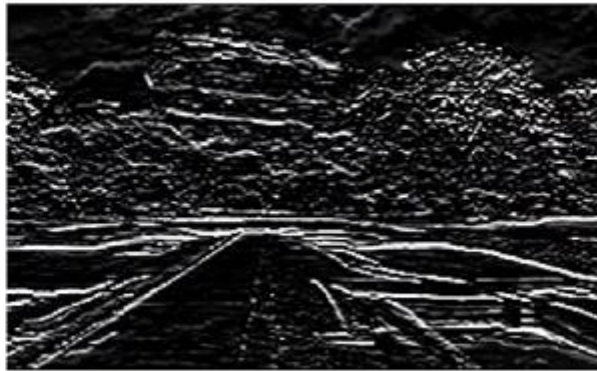
(a) Original



(b) Averaging



(c) Sobel horizontal



(d) Sobel vertical

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Averaging

-1	-2	-1
0	0	0
1	2	1

Sobel vertical

-1	0	1
-2	0	2
-1	0	1

Sobel horizontal

Figure: 3×3 kernels

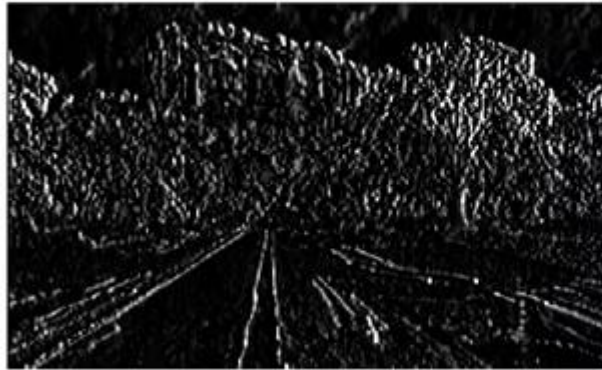
Examples of Effect of Kernel Choices



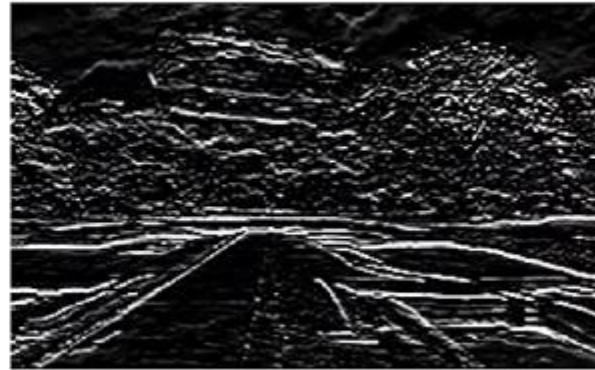
(a) Original



(b) Avegaging



(c) Sobel horizontal



(d) Sobel vertical

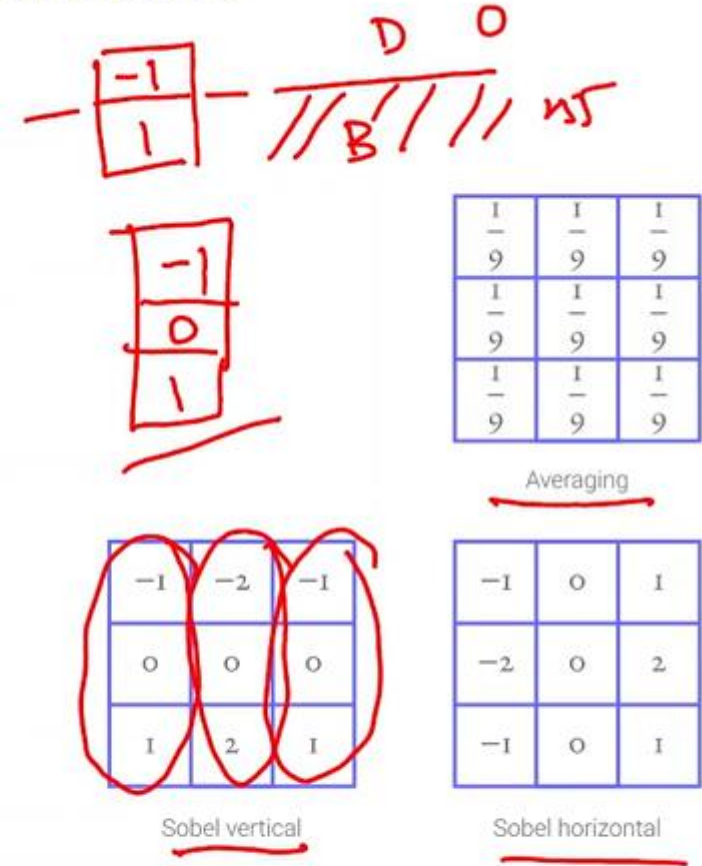


Figure: 3×3 kernels

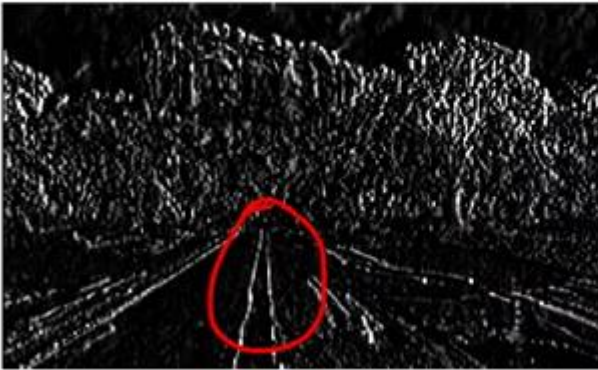
Examples of Effect of Kernel Choices



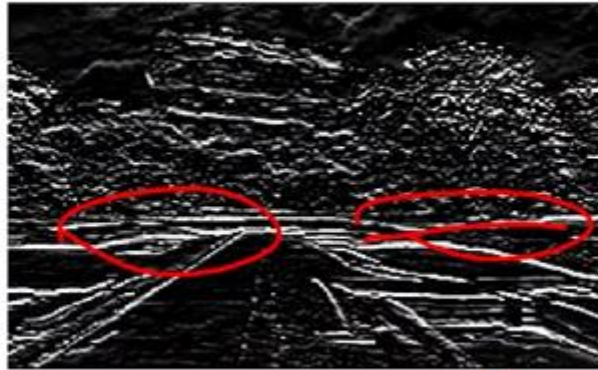
(a) Original



(b) Averaging



(c) Sobel horizontal



(d) Sobel vertical

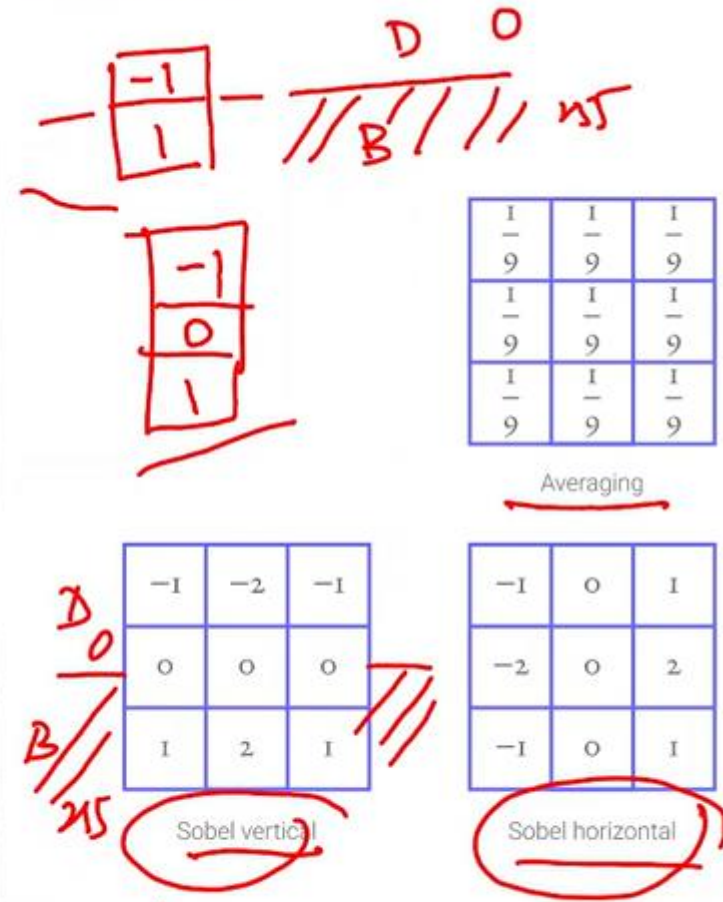


Figure: 3×3 kernels

Reference

- <https://www.youtube.com/watch?v=i1sqNabmmvQ&list=PLELEBz6g6MJanTKBAUJni8FTa8FC8gqGP&index=6>